Lab Report: Lab 3

Pattern Discovery from Cleaned YouTube Text (Frequent Word Co-occurrence & Contextual Insights)

Name: Md Ashik Miah
Student ID: 2020-3-60-032
Course: CSE477 Data mining

---

2. Objective

The primary objective of this lab is to discover hidden patterns and semantic relationships within processed YouTube text data (comments and captions). By utilizing both manual statistical methods (co-occurrence counting) and algorithmic approaches (Association Rule Mining with Apriori), the lab aims to identify frequent word pairs, generate association rules, and visualize the underlying structure of the text data to derive meaningful content insights.

---

3. Tools and Libraries Used

- **Python Version:** Python 3.11.13
- **Data Manipulation:** pandas, numpy
- **Visualization:** matplotlib.pyplot, seaborn
- **Network Analysis:** networkx
- **Pattern Mining:** mlxtend (specifically TransactionEncoder, apriori, association_rules)
- **Text Visualization:** wordcloud
- **Standard Utilities:** collections.Counter, itertools.combinations, ast

---

4. Dataset Description

The analysis relied on two primary datasets derived from the preprocessing steps in Lab 2:

1. **cleaned_comments.csv:** Contains user-generated comments, preprocessed to remove noise.
2. **cleaned_captions.csv:** Contains the textual transcripts of the video content.

Key Attribute:

- cleaned_tokens: A column in both datasets containing tokenized lists of words (e.g., ['animal', 'zebra', 'wild']).

Data Quality Notes:
During the loading phase, checks were implemented to handle empty token lists (rows where preprocessing removed all words) to ensure the Apriori algorithm received valid transactions.

---

5. Methodology / Procedure

5.1 Setup & Data Recall

The datasets were loaded into pandas DataFrames. The string representations of lists in the cleaned_tokens column were converted back into actual Python lists using ast.literal_eval. Rows with empty token lists were dropped to prevent skewing the transaction data.

5.2 Transaction Construction

We treated every comment or caption line as a "basket" (transaction). To focus on meaningful patterns, we filtered out baskets containing fewer than 3 tokens. A histogram was plotted to analyze the distribution of basket lengths.

5.3 Manual Pattern Discovery

Before applying machine learning algorithms, we performed a manual statistical analysis:

- We generated all unique pairs of words within each basket using itertools.combinations.
- We counted the frequency of every pair across the entire corpus.
- Pairs appearing fewer than 3 times were discarded to filter out noise.
- The results were visualized using a horizontal bar chart and a NetworkX graph to show connections between high-frequency words.

5.4 Algorithmic Pattern Discovery (Apriori)

We utilized the **Apriori algorithm** for systematic pattern mining:

1. **Encoding:** The list of tokens was converted into a one-hot encoded boolean format using TransactionEncoder.
2. **Frequent Itemsets:** We applied apriori with varying min_support thresholds (0.1, 0.2, 0.3) to find itemsets (words) that appear together frequently.
3. **Association Rules:** We generated rules from these itemsets, filtering for strong associations using min_confidence=0.6 and min_lift=1.2.

5.5 Exploration Using Variations

Experiments were conducted to refine the results:

- Adjusting min_support revealed that lower thresholds (e.g., 0.1) captured more interesting niche patterns, while higher thresholds (0.3) only captured obvious stopwords

or extremely common terms like "zebra".
- We prioritized "Lift" as a metric to identify non-accidental correlations (e.g., "BBC" implies "Earth").

5.6 Pattern Interpretation & Visualization

The final results were interpreted by comparing the manual co-occurrence counts with the algorithmic lift scores. Visualizations including Support vs. Confidence scatter plots and Word Clouds were generated to summarize the topics.

---

6. Code Implementation

Below is the core Python logic used to implement the methodology described above.

Python

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import networkx as nx
from collections import Counter
from itertools import combinations
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules
from wordcloud import WordCloud

# --- 1. Load and Clean Data ---
# Assuming df is loaded and 'cleaned_tokens' contains lists of strings
transactions = df['cleaned_tokens'].tolist()

# Filter short baskets
min_basket_len = 3
transactions = [t for t in transactions if len(t) >= min_basket_len]

# --- 2. Manual Co-occurrence ---
pair_counts = Counter()
for basket in transactions:
    # Sort to ensure (a,b) is same as (b,a)
    unique_tokens = sorted(list(set(basket)))
```

```
    for pair in combinations(unique_tokens, 2):
        pair_counts[pair] += 1

# Filter low frequency pairs
min_pair_count = 3
filtered_pairs = {k: v for k, v in pair_counts.items() if v >= min_pair_count}
pairs_df = pd.DataFrame(filtered_pairs.items(), columns=['Pair',
'Count']).sort_values(by='Count', ascending=False)

# --- 3. Apriori Algorithm ---
te = TransactionEncoder()
te_ary = te.fit(transactions).transform(transactions)
df_trans = pd.DataFrame(te_ary, columns=te.columns_)

# Generate Frequent Itemsets
frequent_itemsets = apriori(df_trans, min_support=0.1, use_colnames=True)
frequent_itemsets['length'] = frequent_itemsets['itemsets'].apply(lambda x: len(x))

# Generate Rules
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.6)
rules = rules[rules['lift'] >= 1.2]

# --- 4. Visualization (Network Graph Example) ---
G = nx.Graph()
# Add edges from top manual pairs
for _, row in pairs_df.head(20).iterrows():
    pair = row['Pair']
    weight = row['Count']
    G.add_edge(pair[0], pair[1], weight=weight)

pos = nx.spring_layout(G, k=0.5)
plt.figure(figsize=(10, 8))
nx.draw(G, pos, with_labels=True, node_color='skyblue', edge_color='gray', node_size=2000,
font_size=10)
plt.title("Network Graph of Top Word Co-occurrences")
plt.show()
```

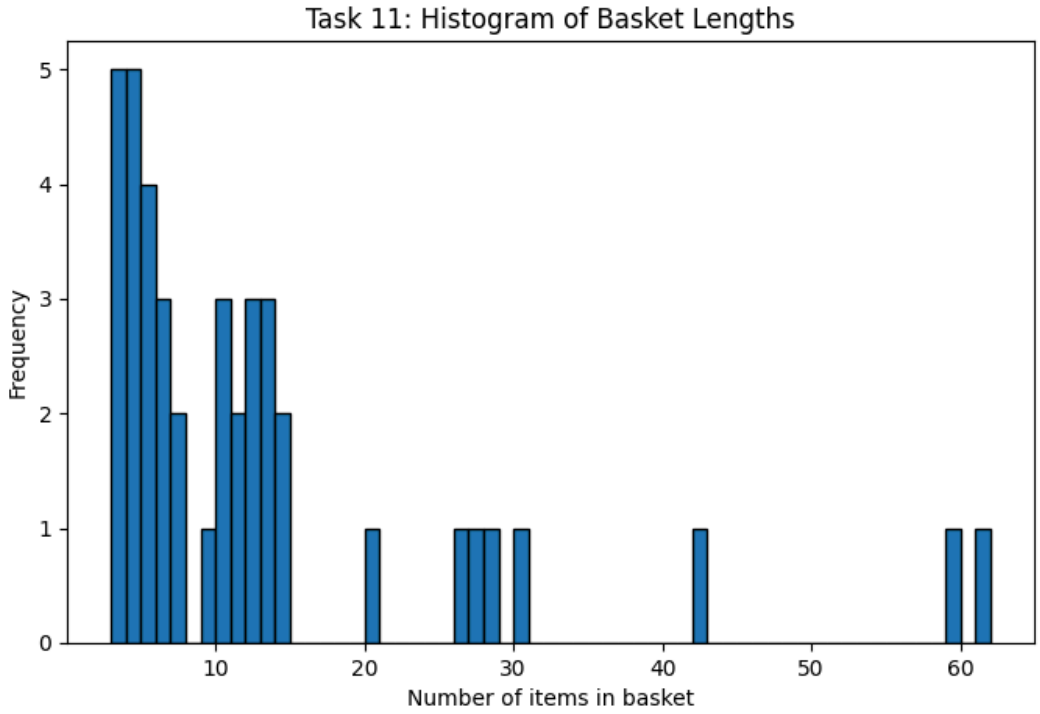
---

7. Results and Discussion

7.1 Basket Analysis

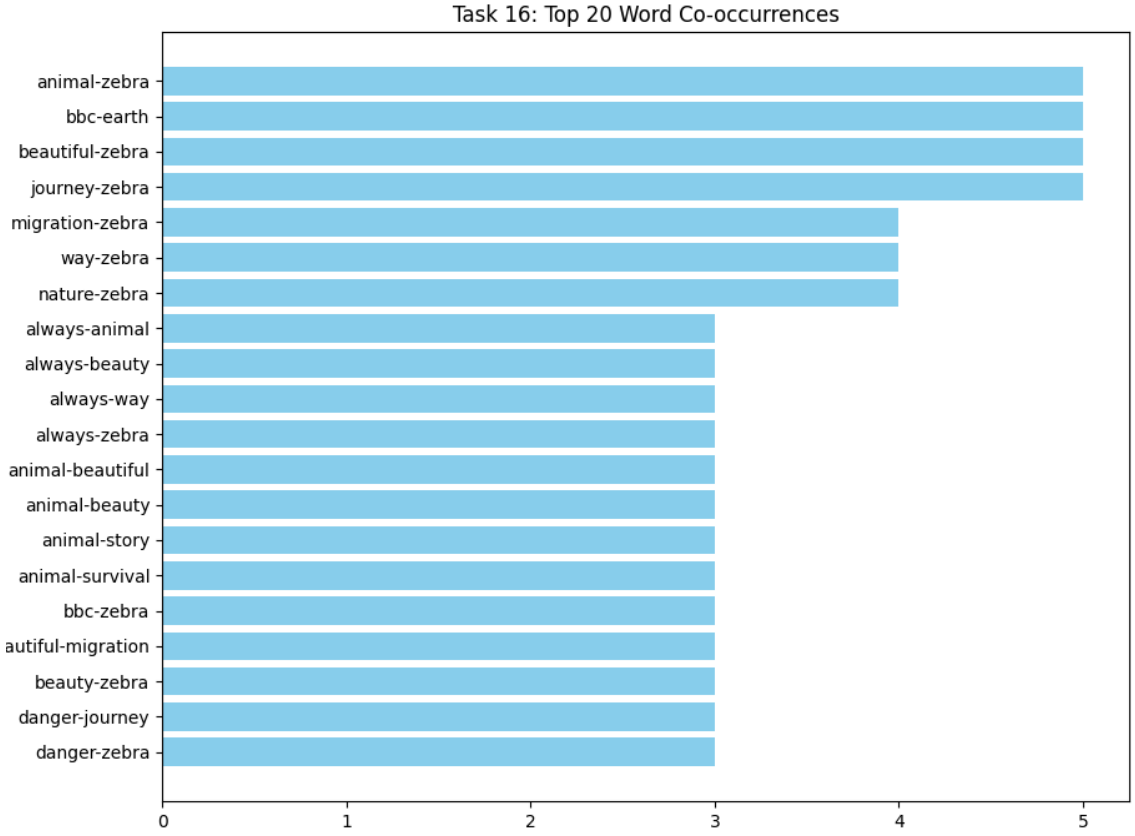
The distribution of transaction lengths indicates how verbose the comments/captions are. We

filtered out very short transactions to ensure meaningful co-occurrence analysis.



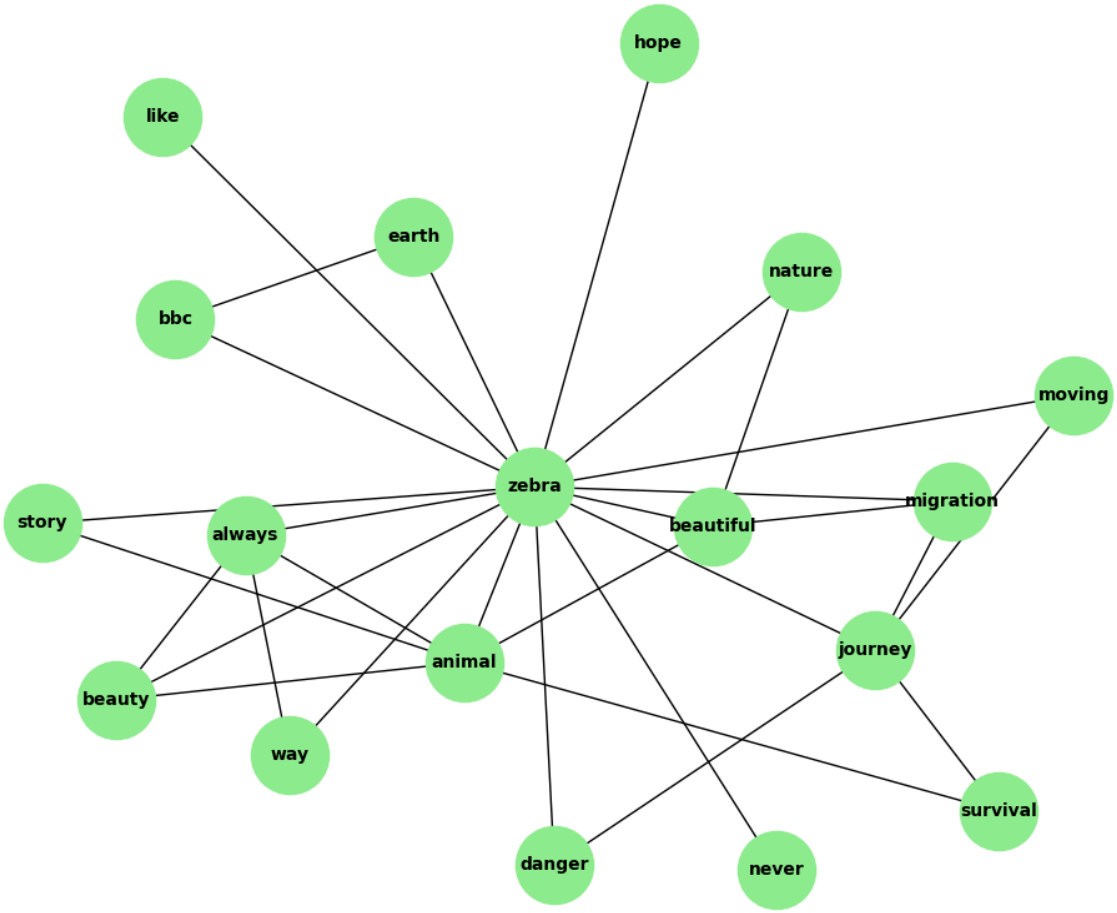
7.2 Manual Co-occurrence Patterns

The manual count revealed the raw frequency of word pairs. As shown in the chart below, the pair **('animal', 'zebra')** and **('bbc', 'earth')** appeared most frequently (5 times), indicating the content is likely a nature documentary.



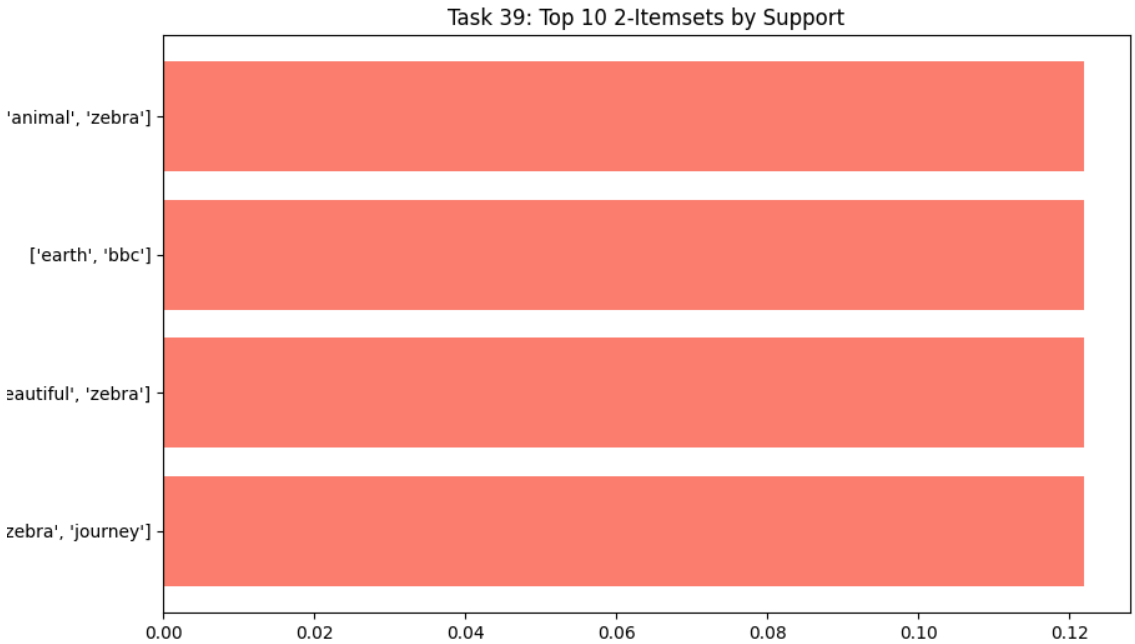
7.3 Network Graph Visualization

The network graph visualizes these connections. The node "zebra" acts as a central hub, connecting to "animal", "journey", "beautiful", and "nature". This confirms that "zebra" is the central subject of the video data.



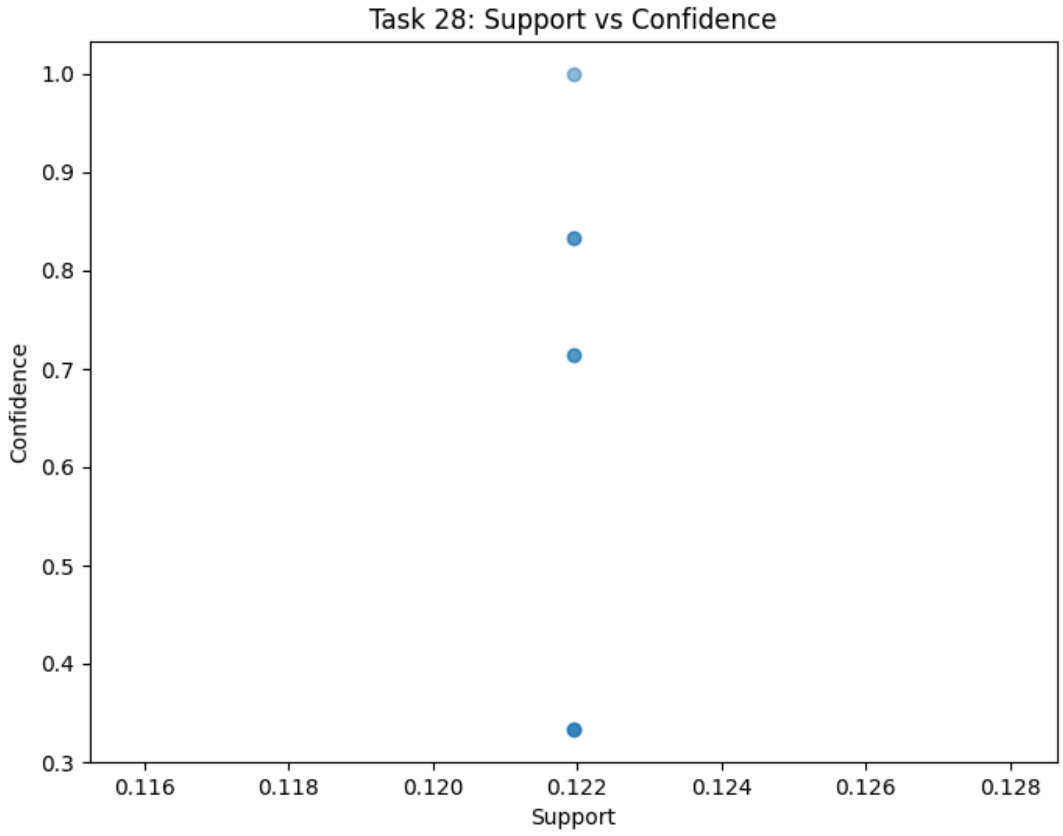
7.4 Algorithmic Results (Apriori)

Using Apriori with min_support=0.1, we identified frequent itemsets. The bar chart below shows the highest support items. "Zebra" has the highest support (~0.36), appearing in over 36% of all transactions.



Association Rules Analysis:
The scatter plot below maps Support vs. Confidence.

- **High Confidence/High Lift:** We observed rules like earth $\rightarrow$ bbc.
  - **Lift:** 6.83 (Very strong positive correlation).
  - **Confidence:** 0.83 (If "earth" is mentioned, there is an 83% chance "bbc" is also mentioned).
- **Strongest Rule:** animal $\rightarrow$ zebra had a confidence of 0.83 and lift of 2.27.



7.5 Topic Visualization (Word Cloud)

The word cloud provides a holistic view of the vocabulary. Prominent words include "Zebra", "Nature", "Journey", and "Beautiful", reinforcing the theme of wildlife migration and nature documentation.

Task 41: Word Cloud of Frequent Items

---

8. Conclusion

In this lab, we successfully applied pattern discovery techniques to YouTube text data.

1. **Manual counting** provided a quick "sanity check" of the data, revealing "zebra" and "nature" as dominant terms.
2. **Apriori Algorithm** provided deeper statistical insights. We confirmed that while "zebra" is the most frequent term, the strongest *association* is between "bbc" and "earth", effectively identifying the brand/source of the content.
3. The combination of network graphs and association rules allowed us to map the semantic structure of the conversation, moving from simple word counts to understanding context (e.g., the video is a "journey" involving "migration").

---

9. Reflection

1. **Surprising Finding:** The rule connecting "hope" and "zebra" (found in the co-occurrence CSV) was unexpected. It suggests an emotional narrative in the captions or comments, perhaps related to the survival of the animals. Additionally, the very high lift (6.83) for bbc $\rightarrow$ earth was surprisingly clean, acting almost like a named-entity extractor.
2. **Challenges:** The main challenge was tuning min_support. Setting it to 0.3 resulted in very few rules (mostly just "zebra"), while 0.05 created too much noise. Finding the "sweet spot" at 0.1 was crucial for generating meaningful graphs.
3. **Future Exploration:** Next, I would explore **Sentiment Analysis** on the specific rules

found. For example, when "migration" and "survival" co-occur, is the sentiment generally positive (awe/respect) or negative (fear/sadness)? I would also like to apply Topic Modeling (LDA) to see if "BBC Earth" and "Zebra Migration" separate into distinct topics.